

Negative Exponents with Variable BasesQuestion Bank - Additional 3.T2 **ANSWER KEY**

Show your work. Circle your final answer.

For questions 1 through 6, write an equivalent expression using only positive exponents. Assume no variable equals 0.

1. b^{-8}

$$\frac{1}{b^8}$$

2. $\frac{1}{m^{-4}}$

$$m^4$$

3. $9x^{-5}$

$$\frac{9}{x^5}$$

4. a^4c^{-3}

$$\frac{a^4}{c^3}$$

5. $\frac{p^2}{q^{-6}}$

$$p^2q^6$$

6. $\frac{4x^{-3}y}{z^{-5}}$

$$\frac{4yz^5}{x^3}$$

For questions 7 and 8, apply the laws of exponents. Write each answer using only positive exponents.

7. Write an equivalent expression.

a. $y^2 \cdot y^{-9}$

$$y^{-7} = \frac{1}{y^7}$$

b. $(2m^{-4})^3$

$$8m^{-12} = \frac{8}{m^{12}}$$

c. $(3x^{-5}y^3)(4x^8y^{-7})$

$$12x^3y^{-4} = \frac{12x^3}{y^4}$$

8. Write an equivalent expression.

a. $\frac{k^3}{k^{10}}$

$$k^{-7} = \frac{1}{k^7}$$

b. $\frac{20a^2b^{-3}}{5a^6b}$

$$4a^{-4}b^{-4} = \frac{4}{a^4b^4}$$

c. $\left(\frac{x^2}{3y^{-2}}\right)^3$

$$\frac{x^6}{27y^{-6}} = \frac{x^6y^6}{27}$$

Working backwards.

9. Find the value of n that makes each statement true.

a. $a^{-3} \cdot a^n = a^4$

$$n = 7$$

b. $\frac{x^n}{x^5} = \frac{1}{x^2}$

$$n = 3$$

$$n - 5 = -2.$$

c. $(y^n)^4 = \frac{1}{y^{20}}$

$$n = -5$$

$$4n = -20.$$

10. Let $x = 4$.

a. Find the value of x^3 and the value of x^{-3} .

64 and 1/64

b. Find the value of $-x^3$. Is it equal to x^{-3} ? Explain.

-64. No — x^{-3} is the reciprocal of x^3 , which is 1/64; $-x^3$ is the opposite of x^3 .

11. Which expression is equivalent to $2y^{-4}$?

☒ A. $\frac{2}{y^4}$

☐ B. $-2y^4$

☐ C. $\frac{1}{16y^4}$

☐ D. $\frac{1}{2y^4}$

A

12. Select ALL of the expressions that are equivalent to $\frac{1}{m^8}$.

☒ A. $m^3 \cdot m^{-11}$

☒ B. m^{-8}

☒ C. $(m^{-4})^2$

☐ D. $(m^{-2})^{-4}$

☐ E. $-m^8$

☒ F. $\frac{m^2}{m^{10}}$

A, B, C and F